Data validation and document retrieval

GROUP QUIZ

Core questions 1-5

Choose one answer for each question. Discuss your reasons as a group before checking with the instructor. Questions 1-20 are the core set; questions 21-25 are optional. These are learning questions, not exam predictions.

1. Kestrel's contract already marks customer_id and customer_name as personal data. Why should it mark the history field as well?

A. The history field is required, and the contract marks every required field as personal data.
B. Its notes carry exact timestamps, and the contract treats every timestamp as personal data.
C. The history field names the staff who wrote each note, and staff names count as customer data.
D. Its notes are free text that can quote the customer, so they can contain any personal details.

2. A colleague suggests that a model infer the case schema from the first record of each export, so that ingestion needs no written contract. Why does Kestrel keep a reviewed, versioned contract instead?

A. An inferred schema can change with the input and miss rules that one record doesn't show.
B. Kestrel's rules don't allow a model to take part in writing a contract, even as a first draft.
C. DynamoDB needs every attribute declared when the table is created, and a model can't supply that.
D. A model can infer field types but not allowed values, and pydantic needs both before it can run.

3. A document's header says Status: current, but its Last reviewed date is more than 365 days before the dataset's reference date. What does ingestion do with the document?

A. It rejects the document, because a document past its review date breaks the document contract.
B. It changes the status to superseded, so that a later retrieval filter can exclude the document.
C. It flags the document as stale in the quality report, and it still loads it into curated/.
D. It loads the document without its metadata file, so that no retrieval filter can select it.

4. A case record has a priority field that contract version 1.0 doesn't name, and its opened_at value is written day first. What does validate_case() return for the record?

A. It returns one reason, for the date, because the model ignores a field that it doesn't declare.
B. It returns two reasons, because the model forbids extra fields and needs the exact UTC form.
C. It returns one reason, for the first rule broken, because validation stops at the first error.
D. It returns no reasons, because it converts the date to UTC and drops the extra field first.

5. Halfway through a run, the team finds that the postcode_area rule rejects KT-10, a service area that Kestrel has just opened. What should happen?

A. Add KT-10 to the pattern in schemas.py, and let the run continue under contract version 1.0.
B. The rule's owner publishes a new contract version, and ingestion runs again under that version.
C. Map each KT-10 record to KT-09, the nearest allowed area, so that the records pass under 1.0.
D. Load the KT-10 records with an empty postcode_area, and ask customer care to correct them later.

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Core questions 6-10

6. A developer proposes loading Kestrel's cases into the knowledge base as well, so that one store serves both kinds of question. What is the main risk?

A. A case lookup would need a filter on case_id, and a managed knowledge base can't filter on equals.
B. The cases would skip the data contract, because the knowledge base ingests files without checks.
C. The cases would lose their history notes, because the knowledge base splits records into chunks.
D. A lookup would become a similarity search, which could return a case of a different customer.

7. The case export holds two records with the same case_id. Their summaries differ, and both pass every other rule. What does ingestion do?

A. It keeps the record that appears later in the file, because a later record is assumed to be newer.
B. It writes both records, because the conditional write stores each version under its own checksum.
C. It accepts the first record and rejects the second, because a person must decide which is right.
D. It counts the second as a duplicate and writes the case once, as it does whenever a key repeats.

8. A case arrives with no owner_team. Most Kestrel cases belong to customer care. What should ingestion do with the case?

A. Fill in customer_care, and record in the quality report that a default value was applied.
B. Reject the case with its reason, because a guessed team could let the wrong staff read it.
C. Load the case with an empty owner_team, so that customer care can correct it in the table.
D. Fill in customer_care, because owner_team holds no personal data and needs no owner's approval.

9. On a second run, put_item fails for one case with AccessDeniedException instead of ConditionalCheckFailedException. How does Kestrel's ingestion code treat the failure?

A. It counts the case as unchanged, because any refused write means the stored item already matches.
B. It retries the write without the condition, so that the case is stored whatever its checksum is.
C. It raises the error, because only a failed condition counts as unchanged, not an access denial.
D. It writes the case to rejected/, because a refused write shows that the record broke the contract.

10. The instructor waits for an ingestion job with a limit of five seconds, and wait_for_ingestion() returns a record with the status TIMED_OUT. What does that record establish?

A. The job failed, because a check that sees no progress before the limit raises an error at once.
B. The job was cancelled, because wait_for stops the AWS job when the deadline passes.
C. The job didn't finish within the chosen limit, and the job itself may still complete later.
D. The sync found nothing to index, because an unchanged sync returns before the first poll.

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Core questions 11-15

11. An engineer traces a citation of kd_07. The manifest row's location matches the citation, but the SHA-256 checksum of the object there differs from the manifest. What can the engineer conclude?

A. The object isn't the version that ingestion accepted, so the citation doesn't establish the source.
B. The object is still valid, because S3 changes an object's checksum when it stores another copy.
C. The knowledge base hasn't synced yet, because a sync rewrites each object's checksum in curated/.
D. The manifest row is out of date, so the engineer should update its checksum to match the object.

12. A quality report shows 40 cases read, 36 accepted and 2 rejected, and it lists one identical duplicate. What should the engineer conclude?

A. The report balances, because an identical duplicate is counted once as read and once as accepted.
B. The run can be used, because 2 rejections out of 40 is within the contract's tolerance for errors.
C. The data quality score is 90 per cent, because 36 of the 40 cases passed every rule in the contract.
D. One case is unaccounted for, so the ingestion code needs investigating before the results are used.

13. A Glue Data Quality evaluation of Kestrel's catalogued cases reports a data quality score of 80 per cent for a ruleset of five rules. What does the score mean?

A. Four of the five rules passed, which says nothing directly about how many cases are valid.
B. Eighty per cent of the cases passed every rule, and the others went to a separate location.
C. Each rule passed for at least 80 per cent of the cases, which meets the default threshold.
D. The service is 80 per cent confident in the data, based on the rules that it recommended.

14. A pipeline loads support tickets into a data lake every hour. It must keep invalid rows out of the lake and send them to a separate location. Where should the DQDL ruleset run?

A. Run it on the lake's table in the Data Catalog, which measures data at rest and can recommend rules.
B. Run it inside the Glue ETL job that loads the rows, which checks data in transit and can route failures.
C. Run it in a Lambda function that EventBridge invokes after each load, which then deletes the failed rows.
D. Run it in Amazon CloudWatch, which evaluates the ruleset against the rows before the job writes them.

15. Kestrel's refund and repair policies change several times a year, and staff must be able to check the source of every answer. Why does the course retrieve passages instead of fine-tuning a model on the documents?

A. Retrieval reads the current passages and their locations at question time, which fine-tuning lacks.
B. Retrieval updates the model's weights at each sync, so the model learns each change within minutes.
C. A fine-tuned model cites the documents that trained it, but code can't check those citations.
D. Fine-tuning means training a new model from scratch, which needs far more data than 25 documents.

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Core questions 16-20

16. A team wants retrieval to match small chunks but return the larger section around each match, so that the model gets more context. What does that require?

A. Use a managed knowledge base with Fixed-size chunking and a large overlap between neighbouring chunks.
B. Use a managed knowledge base with Default chunking, because Default chunking keeps each sentence whole.
C. Change the chunking setting of Kestrel's existing data source, then run a new full sync.
D. Use a customer-managed knowledge base with hierarchical chunking, set when its data source is created.

17. In the comparison, Retrieve returns five scored passages even for the five questions that no Kestrel document answers. What does this show about the scores?

A. The knowledge base found some support for those questions, so the assistant can answer them with care.
B. A score above 0.5 means that the passage answers the question, so the higher passages can be used.
C. The keyword baseline returns nothing for those questions, so it is the safer retriever for them.
D. Retrieve returns its top-ranked passages regardless, so the application decides what is relevant.

18. A developer calls RetrieveAndGenerate on Kestrel's knowledge base and receives "This operation is not supported for managed knowledge bases". Which approach does Kestrel use instead?

A. It sends vectorSearchConfiguration, which lets RetrieveAndGenerate work on a managed knowledge base.
B. It calls AgenticRetrieveStream with generateResponse set to false, which returns a cited answer.
C. It calls Retrieve and writes the answer with its own Converse call, so its code checks each citation.
D. It grants the role bedrock:RetrieveAndGenerate, because the message reports a missing permission.

19. Week 2 will restrict a customer's retrieval to customer documents with a metadata filter on audience. Where should the value that the filter requires come from?

A. It should come from application code, which sets it from the caller's verified role on each call.
B. It should come from the model, which reads the caller's question and chooses a suitable audience.
C. It should come from the caller's message, which says whether the caller is a customer or staff.
D. It should come from each document's metadata file, which records the audience that may read it.

20. On a question that names the router's model, the keyword baseline ranks the supporting document first and the knowledge base ranks it third. On "my broadband keeps dropping", the reverse happens. What explains the difference?

A. The baseline scores passages and the knowledge base scores whole documents, so their units differ.
B. Keyword ranking does well when the question shares the document's words, and hybrid search adds meaning.
C. The knowledge base applied a metadata filter during the comparison, which hid some documents.
D. BM25 gives more weight to long documents, so it ranks a long router guide above a short fault page.

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Optional extra questions 21-25

21. An engineer edits kd_15 in the source folder, but an hour later the assistant still cites the old wording. The data source has no sync schedule. What is the most likely cause?

A. The changed file hasn't been through ingestion and a sync, so the index holds the old chunks.
B. The sync is incremental, so it skips a document whose identifier and file name haven't changed.
C. The model cached its earlier answer, and the cache clears when the next scheduled sync runs.
D. A managed knowledge base syncs daily by default, so the change waits for the next daily run.

22. A client says that the assistant can trust whatever the knowledge base retrieves, because guardrails will be turned on for the model calls. Why is that reasoning unsafe?

A. Guardrails check retrieved passages when the knowledge base is customer-managed, not managed.
B. Guardrails don't apply to passages retrieved from a knowledge base, which can still be wrong.
C. Guardrails check each document only when the knowledge base syncs, so later edits go unchecked.
D. Guardrails check retrieved passages for personal data, so out-of-date rules still pass.

23. An answer cites [kd_15] after each sentence, and kd_15 was among the passages supplied. One sentence states a figure that the passage doesn't contain. What does this case show?

A. The citation check failed, because it should refuse any sentence whose figures differ from the passage.
B. The answer passes the citation check, which checks identifiers, so a review must catch the figure.
C. The answer cites a superseded version of kd_15, because its figure differs from the current passage.
D. The model followed an instruction inside the passage, which the prompt rules should have stopped.

24. A learner's AWS access through their employer will be ready at the start of Week 2. What can the learner do with Exercise 4 today?

A. The learner can do nothing yet, because every cell of Exercise 4 writes to S3 or DynamoDB.
B. Run the whole exercise on the course account, which the instructor opens to learners today.
C. Run the whole exercise locally, because the plan step stands in for the writes to S3 and DynamoDB.
D. Complete validate_case() and put_case_if_changed(), and run the check, which calls no AWS service.

25. A colleague creates a knowledge base whose role can read curated/documents/ only. Its first sync returns COMPLETE, but its statistics show no documents indexed. What should the engineer conclude?

A. The first sync scans the data source, and a second sync then indexes what the first one found.
B. The documents failed the managed parser, because Markdown needs the foundation-model parser.
C. The role must also read curated/metadata/, and COMPLETE doesn't mean that anything was indexed.
D. The sync is still running, because COMPLETE marks the end of scanning, not the end of indexing.